Real-space Visualization of Chern Number for Skyrmions

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1. Research purposes

Tony Skyrme initially predicted skyrmions [1] that are topologically stable field configurations with particle-like properties. Skyrmions are topological magnetic textures [2] that emerge as classical objects in most experimental systems. However, the discovery of skyrmions, characterized by a long lifetime, has generated interest in the quantum field on the nanoscale. Therefore, magnetic skyrmions are envisioned as building blocks of next-generation data storage and processing devices [3], leading to a surge in research activity to identify candidate materials after the early theoretical proposals in magnets [4].

To study topological order in finite systems, Bianco [7] introduced a "topological marker": the local Chern number. In this research, we study the Chern number and local Chern marker of skyrmion.

2. Tight-binding approach

A. Hamiltonian [5]

We begin by considering a tight-binding model for an electron in a square lattice of spin texture $\boldsymbol{n}_i$,

$$H=\sum_{ij}t^{ij}\varphi_i^\dagger\varphi-J\sum\boldsymbol{n}_i\varphi_i^\dagger\boldsymbol{\sigma}\varphi_i$$
$$=\sum_{ij}t^{ij}\varphi_i^\dagger\varphi_j-J\sum(n_{ix}\varphi_i^\dagger\sigma_x\varphi_i+n_{iy}\varphi_i^\dagger\sigma_y\varphi_i+n_{iz}\varphi_i^\dagger\sigma_z\varphi_i), \quad (1)$$

where $\varphi_i^\dagger(\varphi_i)$ is the two-component (spin up and spin down) creation (annihilation) operator at the i site, t^{ij} is the transfer integral between nearest-neighbor sites, J is the Hund's coupling strength between the electron spin and background spin texture, and $\boldsymbol{\sigma}$ denotes the Pauli matrix. Here

$$\sigma_x=\begin{pmatrix}0&1\\1&0\end{pmatrix},\ \sigma_y=\begin{pmatrix}0&-i\\i&0\end{pmatrix},\ \sigma_z=\begin{pmatrix}1&0\\0&-1\end{pmatrix}. \quad (2)$$

For a skyrmion texture, $\boldsymbol{n}_i\equiv\left(\sin\theta_i(r_i)\cos\phi_i(\alpha_i),\sin\theta_i(r_i)\sin\phi_i(\alpha_i)(r_i)\right)$ with $\theta_i=\pi(1-r_i/\lambda)$ and $\phi_i=m\alpha_i+\gamma$.

The Hamiltonian and wavefunction is given by:

$$H=(\varphi_1^\dagger\ \cdots\ \varphi_N^\dagger)\begin{pmatrix}-J\boldsymbol{n}_1\boldsymbol{\sigma}&t_{1,2}&&\\t_{2,1}&\ddots&t_{N-1,N}&\\&t_{N,N-1}&-J\boldsymbol{n}_N\boldsymbol{\sigma}\end{pmatrix}\begin{pmatrix}\varphi_1\\\vdots\\\varphi_N\end{pmatrix}$$
$$=-J[\varphi_1^\dagger\varphi_1\boldsymbol{n}_1\boldsymbol{\sigma}+\varphi_2^\dagger\varphi_1t_{2,1}+\ldots+\varphi_{N-1}^\dagger\varphi_Nt_{N-1,N}+\varphi_N^\dagger\varphi_N\boldsymbol{n}_N\boldsymbol{\sigma}], \quad |\psi\rangle=(c_1\varphi_1^\dagger+\ldots+c_N\varphi_N^\dagger)|0\rangle. \quad (3)$$

According to the Schrödinger equation :

$$H|\psi\rangle=E_n|\psi\rangle$$
$$-J\begin{bmatrix}\varphi_1^\dagger\varphi_1\boldsymbol{n}_1\boldsymbol{\sigma}+\varphi_2^\dagger\varphi_1t_{2,1}+\ldots\\+\varphi_{N-1}^\dagger\varphi_Nt_{N-1,N}+\varphi_N^\dagger\varphi_N\boldsymbol{n}_N\boldsymbol{\sigma}\end{bmatrix}(c_1\varphi_1^\dagger+\ldots+c_N\varphi_N^\dagger)|0\rangle=E_n(c_1\varphi_1^\dagger+\ldots+c_N\varphi_N^\dagger)|0\rangle$$
$$-J\langle0|\varphi_1\begin{bmatrix}\varphi_1^\dagger\varphi_1\boldsymbol{n}_1\boldsymbol{\sigma}+\varphi_2^\dagger\varphi_1t_{2,1}+\ldots\\+\varphi_{N-1}^\dagger\varphi_Nt_{N-1,N}+\varphi_N^\dagger\varphi_N\boldsymbol{n}_N\boldsymbol{\sigma}\end{bmatrix}(c_1\varphi_1^\dagger+\ldots+c_N\varphi_N^\dagger)|0\rangle=\langle0|\varphi_1E_n(c_1\varphi_1^\dagger+\ldots+c_N\varphi_N^\dagger)|0\rangle$$
$$\begin{pmatrix}-J\boldsymbol{n}_1\boldsymbol{\sigma}&t_{1,2}&&\\t_{2,1}&\ddots&t_{N-1,N}&\\&t_{N,N-1}&-J\boldsymbol{n}_N\boldsymbol{\sigma}\end{pmatrix}\begin{pmatrix}c_1\\\vdots\\c_N\end{pmatrix}=E_n\begin{pmatrix}c_1\\\vdots\\c_N\end{pmatrix} \quad (4)$$

By solving this eigenvalue problem, we can obtain the eigenvalues and the eigenstate.

B. In periodic boundary condition [6]

We rewrite the Hamiltonian in the momentum space

$$H_{ij}^{\mathbf{k}}=\langle\chi_i^{\mathbf{k}}|H|\chi_j^{\mathbf{k}}\rangle=\sum_{\mathbf{R}_i,\mathbf{R}_j}e^{i\mathbf{k}\cdot(\mathbf{R}_i-\mathbf{R}_j)}\langle\varphi_{\mathbf{R}i}|H|\varphi_{\mathbf{R}j}\rangle, \quad (5)$$

and Bloch functions is given by

$$|\psi_{n\mathbf{k}}\rangle=\sum_jC_j^{n\mathbf{k}}|\chi_j^{\mathbf{k}}\rangle$$
$$=\sum_{\mathbf{R}_j}C_j^{n\mathbf{k}}e^{i\mathbf{k}\cdot(\mathbf{R}_j)}|\varphi_{\mathbf{R}j}\rangle,$$
$$\text{where }|\chi_j^{\mathbf{k}}\rangle=\sum_{\mathbf{R}_j}e^{i\mathbf{k}\cdot(\mathbf{R}_j)}|\varphi_{\mathbf{R}j}\rangle. \quad (6)$$

The Chern number [6] assigned to the nth band is defined by

$$c_n=\frac{1}{2\pi\mathrm{i}}\int_{T^2}\mathrm{d}^2kF_{12}(k), \quad (7)$$

where the Berry connection $A_\mu(k)$ $(\mu=1,2)$ and the associated field strength $F_{12}(k)$ are given by

$$A_\mu(k)=\langle n(k)|\partial_\mu|n(k)\rangle,$$
$$F_{12}(k)=\partial_1A_2(k)-\partial_2A_1(k), \quad (8)$$

with $|n(k)\rangle$ being a normalized wave function of the nth Bloch band such that $H(k)|n(k)\rangle=E_n(k)|n(k)\rangle$.

C. Local properties

Local Chern marker [7]

The topological phases of finite-size systems can be characterized by the topological marker with the local Chern number as [7]

$$\mathfrak{C}(\mathbf{r})=-2\pi i\int d\mathbf{r}'\left[\tilde{X}(\mathbf{r},\mathbf{r}')\tilde{Y}(\mathbf{r}',\mathbf{r})-\tilde{Y}(\mathbf{r},\mathbf{r}')\tilde{X}(\mathbf{r}',\mathbf{r})\right], \quad (9)$$

where $\tilde{X}$ is the projected x coordinate:

$$\tilde{X}(\mathbf{r},\mathbf{r}')=\int d\mathbf{r}''P(\mathbf{r},\mathbf{r}'')x''P(\mathbf{r}'',\mathbf{r}'), \quad (10)$$

and similarly $\tilde{Y}$. P is the groundstate projector:

$$P=\frac{A_c}{(2\pi)^2}\sum_{n=1}^{N}\int_{\mathrm{BZ}}d\mathbf{k}|\psi_{n\mathbf{k}}\rangle\langle\psi_{n\mathbf{k}}|. \quad (11)$$

Single-point local marker [8]

The usual local Chern marker can only be applied to open boundary conditions. In systems with periodic boundary conditions, we need to compute the Single-point local marker to obtain the local Chern marker. A symmetric Single-point local marker defined at the lattice site $\mathbf{r}$[8],

$$C^{(\mathrm{sym})}(\mathbf{r})=-\frac{1}{4\pi}\mathrm{Im}\langle\mathbf{r}|\bar{P}|\mathbf{r}\rangle$$
$$=-\frac{1}{8\pi}\mathrm{Im}\langle\mathbf{r}|([\mathcal{P}_{\mathbf{b}_1},\mathcal{P}_{\mathbf{b}_2}]+[\mathcal{P}_{-\mathbf{b}_1},\mathcal{P}_{-\mathbf{b}_2}]$$
$$-[\mathcal{P}_{-\mathbf{b}_1},\mathcal{P}_{\mathbf{b}_2}]-[\mathcal{P}_{\mathbf{b}_1},\mathcal{P}_{-\mathbf{b}_2}])\mathcal{P}_\Gamma|\mathbf{r}\rangle. \quad (12)$$

Where $\mathbf{b}_{1,2}$ are the reciprocal lattice vectors and the projectors $\mathcal{P}_{b_j}$ and $\mathcal{P}_\Gamma$ as

$$\mathcal{P}_{\mathbf{b}_i}=\sum_{n=1}^{N_{\mathrm{cell}}}|\tilde{u}_{n\mathbf{b}_i}\rangle\langle\tilde{u}_{n\mathbf{b}_i}|,\qquad\mathcal{P}_\Gamma=\sum_{n=1}^{N_{\mathrm{cell}}}|u_{n\Gamma}\rangle\langle u_{n\Gamma}|. \quad (13)$$

$|\tilde{u}_{n\mathbf{b}_j}\rangle$ are the "dual" states of the Hamiltonian eigenstates $|u_{n\Gamma}\rangle$ as

$$|\tilde{u}_{n\mathbf{b}_i}\rangle=\sum_{m=1}^{N_{\mathrm{cell}}}S_{nm}^{-1}(\mathbf{b}_j)e^{-i\mathbf{b}_j\cdot\mathbf{r}}|u_{m\Gamma}\rangle, \quad (14)$$

$S_{nm}(\mathbf{b}_j)$ is the overlap matrix as

$$S_{nm}(\mathbf{b}_j)=\langle u_{n\Gamma}|e^{-i\mathbf{b}_j\cdot\mathbf{r}}|u_{m\Gamma}\rangle \quad (15)$$

3. Results and Discussion

Figure 1(a) shows a view of a unit cell of 6×6 skyrmion crystal (SKX) [9], with arrows depicting the spin direction. Figure 1(b) shows the band structure obtained by numerically diagonalizing the effective Hamiltonian in the unit cell with the size 4 × 4 in the presence of the skyrmion crystal. We calculated the Chern number by the Fukui-Hatsugai method [6] on each occupied energy band. Interestingly, the 10th band of Figure 2 has an extremely large Chern number of $C_{10}=5$, as shown in Figure 1(c). Summing all the Chern number of the energy bands in all occupied states is equal to 0.

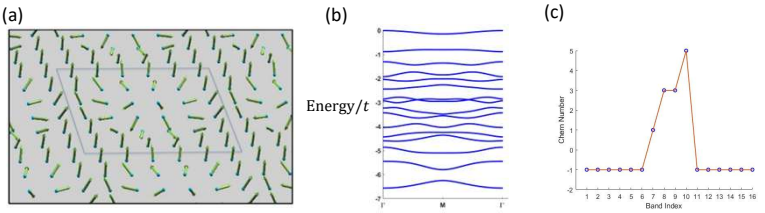


Figure 1. (a) A view of a unit cell of 6×6 SkX. (b) The band structure in the presence of SkX. The parameters $J/t=3$. (c) The band index dependence of Chern number. The parameters $J/t=3$.

Figures 2 (a) and (b) show a schematic of a 3×3 Néel skyrmion lattice and a unit cell with a size 14×14 configuration. (c), (d) show their local Chern marker $\mathfrak{C}(\mathbf{r})$. The local Chern marker averaged over the bulk area gives the Chern number equal to 0.

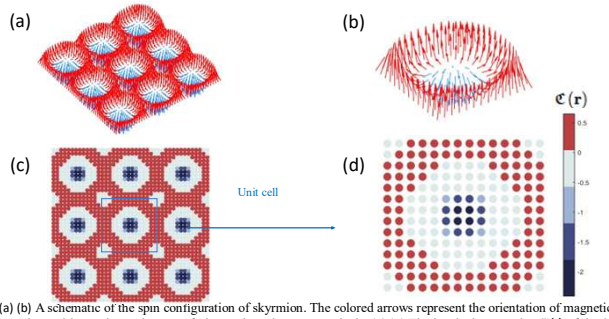


Figure 2. (a) (b) A schematic of the spin configuration of skyrmion. The colored arrows represent the orientation of magnetic moments. Red and blue represent the positive and negative out-of-plane orientations, respectively. (c) (d) The local Chern marker $\mathfrak{C}(r)$ of the 3×3 skyrmion lattice and a unit cell with the size 14×14 configuration. The skyrmion radius is $\lambda=7$. The parameters $J/t=3$.

4. Γ-point implementation of local marker in DFT code

The OpenMX is based on density functional theories (DFT), the norm-conserving pseudopotentials, and local pseudo-atomic basis functions. The Kohn-Sham (KS) Bloch functions ψ_μ are expanded in a form of linear combination of pseudo-atomic basis functions (LCPAO) $\phi_{i\alpha}$ centered on site τ_i by

$$\psi_{\sigma\mu}^{(\mathbf{k})}(\mathbf{r})=\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}u_{\sigma\mu}^{(\mathbf{k})}(\mathbf{r})$$
$$=\frac{1}{\sqrt{N}}\sum_{\mathbf{n}}^{N}\mathrm{e}^{\mathrm{i}\mathbf{R}_{\mathbf{n}}\cdot\mathbf{k}}\sum_{i\alpha}c_{\sigma\mu,i\alpha}^{(\mathbf{k})}\phi_{i\alpha}(\mathbf{r}-\tau_i-\mathbf{R}_{\mathbf{n}}), \quad (16)$$

where c and ϕ are an expansion coefficient and pseudo-atomic function, $\mathbf{R}_{\mathbf{n}}$ a lattice vector, i a site index, σ (↑ or ↓) spin index, $\alpha\equiv(plm)$ an organized orbital index with a multiplicity index p, an angular momentum quantum number l, and a magnetic quantum number m.

In the OpenMX, the formulas of how to implement the Single-point local marker are described as follows

$$C^{(\mathrm{sym})}(\mathbf{r})=-\frac{1}{8\pi}\mathrm{Im}\langle\mathbf{r}|([\mathcal{P}_{\mathbf{b}_1},\mathcal{P}_{\mathbf{b}_2}]+[\mathcal{P}_{-\mathbf{b}_1},\mathcal{P}_{-\mathbf{b}_2}]-[\mathcal{P}_{-\mathbf{b}_1},\mathcal{P}_{\mathbf{b}_2}]-[\mathcal{P}_{\mathbf{b}_1},\mathcal{P}_{-\mathbf{b}_2}])\mathcal{P}_\Gamma|\mathbf{r}\rangle$$
$$=-\frac{1}{8\pi}\mathrm{Im}\left\{\begin{array}{l}[\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_\Gamma|\mathbf{r}\rangle-\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_\Gamma|\mathbf{r}\rangle]+[\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_\Gamma|\mathbf{r}\rangle-\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_\Gamma|\mathbf{r}\rangle]\\-[\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_\Gamma|\mathbf{r}\rangle-\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_\Gamma|\mathbf{r}\rangle]-[\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_\Gamma|\mathbf{r}\rangle-\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_1}\mathcal{P}_{\mathbf{b}_2}\mathcal{P}_\Gamma|\mathbf{r}\rangle]\end{array}\right\} \quad (17)$$

$$\langle\mathbf{r}|\mathcal{P}_{\mathbf{b}_i}\mathcal{P}_{\mathbf{b}_j}\mathcal{P}_\Gamma|\mathbf{r}\rangle=\sum_{l,m,n}\langle\mathbf{r}|\tilde{u}_{l\mathbf{b}_i}\rangle\langle\tilde{u}_{l\mathbf{b}_i}|\tilde{u}_{m\mathbf{b}_j}\rangle\langle\tilde{u}_{m\mathbf{b}_j}|u_{n\Gamma}\rangle\langle u_{n\Gamma}|$$
$$=\sum_{l,m}\langle\mathbf{r}|\tilde{u}_{l\mathbf{b}_i}\rangle\langle\tilde{u}_{l\mathbf{b}_i}|\tilde{u}_{m\mathbf{b}_j}\rangle\langle u_{m\Gamma}|\mathbf{r}\rangle,\ n=m$$
$$=\sum_{l,m}\langle\phi_{i\alpha}|\tilde{u}_{l\mathbf{b}_i}\rangle\langle\tilde{u}_{l\mathbf{b}_i}|\tilde{u}_{m\mathbf{b}_j}\rangle\langle u_{m\Gamma}|\phi_{i\alpha}\rangle,\ n=m \quad (18)$$

$$|\tilde{u}_{\mathbf{b}_i}\rangle=\sum_{m'}^{N_{\mathrm{cell}}}|u_{m'\Gamma}\rangle S_{m'1}^{-1}(\mathbf{b}_i)$$
$$\langle\phi_{i\alpha}|\tilde{u}_{\mathbf{b}_i}\rangle=\sum_{m'}^{N_{\mathrm{cell}}}\langle\phi_{i\alpha}|u_{m'\Gamma}\rangle S_{m'1}^{-1}(\mathbf{b}_i)$$
$$=\sum_{m'}^{N_{\mathrm{cell}}}\langle\phi_{i\alpha}|\phi_{j\beta}\rangle c_{j\beta}^{m'\Gamma}S_{m'1}^{-1}(\mathbf{b}_i)$$
$$=\sum_{m'}^{N_{\mathrm{cell}}}\sum_{j\beta}\sum_{\mathbf{R}_n}S_{i\alpha,j\beta}^{\mathbf{R}_n}c_{j\beta}^{m'\Gamma}S_{m'1}^{-1}$$
Where $S_{i\alpha,j\beta}^{\mathbf{R}_n}=\langle\phi_{i\alpha}|\phi_{j\beta}\rangle$ and $S_{m'1}^{-1}(\mathbf{b}_i)=(\langle u_{m'\Gamma}|e^{-i\mathbf{b}_i\cdot\mathbf{r}}|u_{1\Gamma}\rangle)^{-1}$

$$\langle\tilde{u}_{\mathbf{b}_i}|\tilde{u}_{m\mathbf{b}_j}\rangle=\sum_{l'm'}^{N_{\mathrm{cell}}}S_{l'1}^{-1*}(\mathbf{b}_i)\langle u_{l'\Gamma}|u_{m'\Gamma}\rangle S_{m'1}^{-1}(\mathbf{b}_2)$$
$$=\sum_{l'}^{N_{\mathrm{cell}}}S_{l'1}^{-1*}(\mathbf{b}_2)S_{l'm}^{-1}(\mathbf{b}_2),\ l'=m'$$
Here $S_{l'1}^{-1}(\mathbf{b}_2)=(\langle u_{l'\Gamma}|e^{-i\mathbf{b}_2\cdot\mathbf{r}}|u_{1\Gamma}\rangle)^{-1}$ and $S_{l'm}^{-1}(\mathbf{b}_2)=(\langle u_{l'\Gamma}|e^{-i\mathbf{b}_2\cdot\mathbf{r}}|u_{m\Gamma}\rangle)^{-1}$

$$\langle u_{m\Gamma}|\phi_{i\alpha}\rangle=\langle\psi_{m\Gamma}|\phi_{i\alpha}\rangle$$
$$(\langle\psi_{m\Gamma}|\phi_{i\alpha}\rangle)^*=\langle\phi_{i\alpha}|\psi_{m\Gamma}\rangle$$
$$=\langle\phi_{i\alpha}|\sum_{j\beta}\sum_{\mathbf{R}_n}|\phi_{j\beta}^{\mathbf{R}_n}\rangle c_{j\beta}^{m'\Gamma}$$
$$=\sum_{j\beta}\sum_{\mathbf{R}_n}\langle\phi_{i\alpha}|\phi_{j\beta}^{\mathbf{R}_n}\rangle c_{j\beta}^{m'\Gamma}$$
$$=\sum_{j\beta}\sum_{\mathbf{R}_n}S_{i\alpha,j\beta}^{\mathbf{R}_n}c_{j\beta}^{m\Gamma} \quad (19)$$

5. Summary and prospect

We obtained the band structure in periodic conditions and calculated each occupied energy band Chern number. Then, we compute the local Chern marker $\mathfrak{C}(\mathbf{r})$ of the 3×3 skyrmion lattice and unit cell with the size 14×14 configuration. Furthermore, we derive formulas for the calculation of Single-point local marker in DFT.

We hope this computational method can be applied to realistic model computation [10].

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